

frequency is measured in Hertz (Hz). Human ears can recognise frequencies spanning 20Hz to 20,000Hz at most. This range makes up the entire audio spectrum. But our ears are most receptive to mid-range frequencies only.

Low midrange, that is, 250Hz to 500Hz, covers the lower-order elements of songs like bass and any lower-stringed instruments.

Midrange, that is, 500Hz to 2kHz, controls loudness and prominence of an instrument among other instruments in a song.

Upper midrange, that is, 2kHz to 4kHz, determines audio levels of instruments like drums or guitars to provide a sonic boost where needed.

Multiple connectivity options

Bluetooth allows sending music from tablets, smartphones or laptops wirelessly to the soundbar speaker. Some soundbars also support near field communication (NFC), which is a quick way to make an initial Bluetooth connection. Others have two-way Bluetooth feature, which lets you send music from the soundbar to Bluetooth-enabled speakers or headphones.

Models with built-in Wi-Fi let you access online music services and stream movies or TV shows directly from the soundbar. Some models include an Ethernet jack for a wired connection to your home network.

Soundbars compatible with digital assistants, such as Google Assistant and Amazon Alexa, allow you to control them with just your voice. You can turn up the volume, skip to the next track or stop the music, all without lifting a finger. You can also control the volume and power it on/off with a single remote.

Optical digital audio inputs and outputs support Stereo, Dolby Digital and DTS sounds, while HDMI cables support Stereo, Dolby Digital, DTS, DTS:X, Dolby Digital Plus and Dolby Atmos. With up to two HDMI input and one output connectivity, multiple devices can be connected to the soundbar. USB inputs are only included for firmware updates.

HDMI ARC (Audio Return Channel) is a protocol that appears on

most of the new soundbars with HDMI connection. This allows the TV and soundbar to exchange information, including the ability to route video to the TV, and sound from the TV, all over a single connection.

How soundbars are different from other sound systems

Sound bases have speaker arrays like soundbars, but most have built-in sub-woofers rather than separate ones. Due to the lack of separate sub-woofers, these struggle to produce the same kind of deep bass as traditional soundbars with sub-woofers.

Soundbars use Wi-Fi to connect to standalone wireless speakers. These do a better stereo channel separation because the speakers are further apart.

A home cinema system gives true surround sound, but a soundbar offers improved audio with virtual surround sound, without the need for wires and multiple speakers.

Real home theatre sound comes in many variations, but typically, home theatre sound requires a separate amplifier and at least five speakers—one centre, one left, one right and two rear—and a sub-woofer, which usually costs more than a soundbar alone. If you have space and the budget, you might want to consider authentic home theatre sound.

Most shows and movies these days offer five or more different audio channels or speakers embedded digitally—centre, left, right and two in the rear (left rear and right rear). What makes a soundbar different from a typical home theatre setup is that all channels/speakers are contained in one unit.

Home theatre usually has a five-channel system. A seven-channel soundbar has seven speakers. Essentially, a seven-channel system is the same as a five-channel system with a bonus split surround and rear channel information in four channels.

How to select a soundbar

Soundbars are identified through numbers like 2.0, 3.1 or 5.1.4. Here, the first number refers to the number of channels or speakers, and the second number tells whether there

is a sub-woofer (1) or not (0). Two channels means two drivers, left and right, while three means left, right and centre; five means added channels for rear or surround sound speakers.

The third number, where applicable, signifies whether the soundbar supports Dolby Atmos surround sound. And this it also refers to the number of dedicated drivers that fire upwards at the ceiling, bouncing sound down to create an enveloping effect.

Some surround sound speakers are built to be mounted on the ceiling, though not in a soundbar setup. Atmos is currently the most popular surround sound technology, capable of processing up to 128 distinct objects in a given scene.

If you simply want to enhance your TV sound, a soundbar with 2.1 channels (two front channels and a separate sub-woofer) could be enough. The cheaper the model you choose, the more basic connections are likely to be made.

Most speakers also come with a remote control and digital LED display, bringing added comfort and convenience to usage. Newer soundbars come with good sound quality, good battery life and superb design finish, and use latest technologies like intelligent chip control, which prevents overheating and promotes stabilised charging. Soundbars like Sound Blast launched by Toreto, in addition to above-mentioned features, also allow playing chart-busters via FM Radio.

Some soundbars can be connected to a wireless multi-room system where different speakers are connected around the house. These speakers are all controlled from a single app and can be used to play different or the same music.

Active means the soundbar comes with built-in amplifiers that power everything, as well as channel processors that separate left, right and centre speakers in the soundbar. So, no extra receiver is needed. Whereas passive requires an external speaker, sub-woofer and amplifiers. However, if you are looking to upgrade your TV's audio, go with an active soundbar. **EFY**